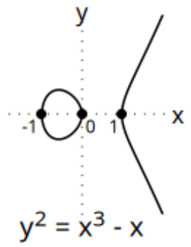
Fermat's last theorem for n=4
(any field K has been replaced with $\mathbb{Q}$)

**Def** * Elliptic curve over $\mathbb{Q}$ is equation of form:
$y^2=ax^3+bx^2+cx+d$ (a,b,c,d∈Q, a≠0)
where RHS polynomial has no multiple root

**Def** For $a=\frac{m}{n}$ with m,n∈ℕ, (m,n)=1
define height H(a):=max(|m|, |n|)

(x,y) on elliptic curve
⇒ $(\frac{-1}{x},\frac{y}{x^2})$ on as well,
H(x)=H(-1/x)

**Lem 1.5** For distinct a,b,c∈$\mathbb{Q}$ define B, $\tilde{C}$ and C
$B=\{(u,v,w)\in\mathbb{Q}\times\mathbb{Q}\times\mathbb{Q} | u^2+a=v^2+b=w^2+c\}$
$\tilde{C}=\{(x,y)\in\mathbb{Q}\times\mathbb{Q} | y^2=(x-a)(x-b)(x-c)\}$.
$C=\{(x,y)\in\mathbb{Q}\times\mathbb{Q} | y^2=(x-a)(x-b)(x-c), y\neq 0\}$.
(1) ∃ mutually inverse maps f:B→C, g:C→B
$f(u,v,w)=(u^2+a+uv+vw+wu, (u+v)(v+w)(w+u)$,
$g(x,y)=\left(\frac{1}{2y}((x-a)^2-(b-a)(c-a)\right.$,
$\left.\frac{1}{2y}((x-b)^2-(a-b)(c-b)), \frac{1}{2y}((x-c)^2-(a-c)(b-c))\right)$
(2) ∃ map h:B→$\tilde{C}$, $h(u,v,w)=(u^2+a,uvw)$




$i:(x,y,z)\mapsto\left(\frac{y-x}{2},\frac{z}{2},\frac{x+y}{2}\right)$
$j:(u,v,w)\mapsto(w-u,w+u,2v)$

$\{ i:A_d\to B_d,\ j:B_d\to A_d$ a=d, b=0, c=-d, $\{ f:B_d\to C_d,\ g:C_d\to B_d$

**Rem 1.6** h∘g:C→$\tilde{C}$ is called multiplication-by-2 map
for elliptic curve $y^2=(x-a)(x-b)(x-c)$; image of h∘g is
$\{(x,y)\in\mathbb{Q}\times\mathbb{Q} | y^2=(x-a)(x-b)(x-c)$, x-a, x-b, x-c are □s in $\mathbb{Q}\}$
because:
$(x',y')=h(u,v,w) \Rightarrow x'=u^2+a \Rightarrow x'-a$ is □ in $\mathbb{Q}$
from definition of B (image of g): $u^2+a=v^2+b=w^2+c$
$x'-b=(v^2+b)-b=v^2$ is □ in $\mathbb{Q}$
$x'-c=(w^2+c)-c=w^2$ is □ in $\mathbb{Q}$

a=1, b=0, c=-1

**Lem 1.4** For d∈$\mathbb{Q}$ define:
$A_d=\{(x,y,z)\in\mathbb{Q}\times\mathbb{Q}\times\mathbb{Q} | x^2+y^2=z^2, ½xy=d\}$
$B_d=\{(u,v,w)\in\mathbb{Q}\times\mathbb{Q}\times\mathbb{Q} | u^2+d=v^2,\ v^2+d=w^2\}$
$C_d=\{(x,y)\in\mathbb{Q}\times\mathbb{Q} | y^2=x^3-d^2x,\ y\neq 0\}$.
∃ bijections between any 2 of $A_d$, $B_d$, $C_d$

$\{$ (i) $\Rightarrow A_d\neq\varnothing$
(ii) $\Rightarrow B_d\neq\varnothing$
(iii) $\Rightarrow C_d\neq\varnothing$

**(i)** Let $(x_0,y_0)$ on $y^2=x^3-x$ with $y_0\neq 0$
and minimal $H(x_0)$; assume $1<x_0$

**Lem 1.3** d∈$\mathbb{Q}$, these are equivalent:
(i) ∃ rational number sides **right** △ of area d
(ii) ∃ 3 □s of rational numbers that form
arithmetic progression of difference d
(iii) ∃ rational solution to $y^2=x^3-d^2x$
other than (x,y)=(0,0) and (±d,0)

¬(i) ⟺ d=1

**Prop 0.13** The area of a right △
whose sides are integers is not a □

¬(iii) ⟺ d=1

**Prop 1.2** The only rational solutions
to $y^2=x^3-x$ are (x,y)=(0,0) and (±1,0)

**(ii)** $x_0-1$, $x_0$ and $x_0+1$ are all squares
⇒ $\exists(x_1,y_1)\in C$: $h\circ g(x_1,y_1)=(x_0,y_0)$

1.1 sol · $z^2/y^6$, ↯:
$0\neq\left(\frac{x^2z}{y^3}\right)^2=\left(\frac{z^2}{y^2}\right)^3-\frac{z^2}{y^2}$

**Prop 1.1** ∄ solution (x,y,z) to
$x^4+y^4=z^4$ satisfying xyz≠0

**(iii)** $H(x_1)<H(x_0)$, ↯